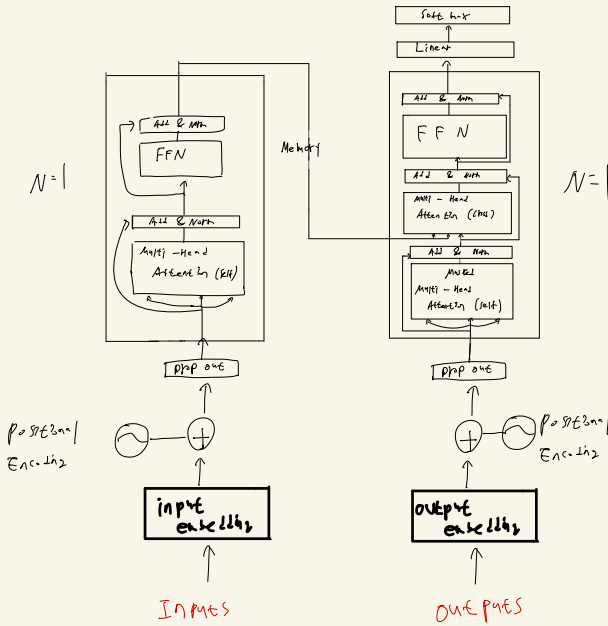


Transformer's Update in Small D_{cho}

$$N=1, \quad h=2, \quad d_{\text{mod}}=4, \quad \text{Vocst-Size}=16, \quad d_{\text{fft}}=8, \quad \text{par}78 = (\text{"Ihm" in base "hexa"}, \text{"42" in base "hexa"}).$$


```
Encoder tokens: ['I', ' ', 'n', ' ', 'the', ' ', 'best', ' ', 'boxer', ' ', ' ', '<EOS>']
Decoder input tokens: ['<BOS>', '나는', '최고', '의', '복', '을', '써', '다', '.']
Labels: ['나는', '최고', '의', '복', '을', '써', '다', '.', '<EOS>']
```

```
source(encoder_input):
    tensor([[[[ 0.0000, 1.1111, 0.0000, 1.1111],
               [ 0.9150, 0.0000, 0.0111, 1.1111],
               [ 1.0183, -0.4624, 0.0222, 1.1100],
               [ 1.1506, -1.1000, 0.0333, 1.1000],
               [ 0.9150, -0.7283, 0.0444, 1.1000],
               [ 1.0055, 0.0000, 0.0555, 1.0977],
               [ 0.3185, 1.0669, 0.0666, 0.0000],
               [ 0.7280, 0.0377, 0.0777, 1.0844]]]]) device='mps', grad_fn=NullBackward0)

target(decoder_input):
    tensor([[[[ 0.0000, 1.1111, 0.0000, 1.1111],
               [ 0.9150, 0.0000, 0.0111, 1.1111],
               [ 1.0183, -0.4624, 0.0222, 1.1100],
               [ 1.1506, -1.1000, 0.0333, 1.1000],
               [ 0.9150, -0.7283, 0.0444, 1.1000],
               [ 1.0055, 0.0000, 0.0555, 1.0977],
               [ 0.3185, 1.0669, 0.0666, 0.0000],
               [ 0.7280, 0.0377, 0.0777, 1.0844]]]]) device='mps', grad_fn=NullBackward0)
```

$$X = \sqrt{\text{Var}(X)} \cdot \text{Erkennung}(X) + \mu$$

$$\in \mathbb{R}^{8 \times 7} \text{ (Lernen \& Train)}$$

$$\kappa_K = d_{K, H}/h = F/2 = 2 \quad W^a, W^k, W^v \in \mathbb{R}^{4 \times 1 (d_{K, H}/h)}$$

Encoder

$Q = x^s w^s, K = x^s w^k, V = x^s w^v, \in \mathbb{R}^{(\text{embedding} \times \text{head})}$

$\downarrow$

$\text{softmax}(h \times 2)$

```

graph LR
    head1[head1] --> head2[head2]
    head2 --> null(( ))
    style null stroke-dasharray: 5 5
  
```

$$\boxed{\text{S.1.10.1}}: S_1 = \frac{Q_1 \cdot K_1^T}{\sqrt{K_1}} \in \mathbb{R}^{\text{columns: } K_1, \text{ rows: } Q_1}, A_i = \text{Jacobian}(S_i), O_i = A_i^T V_i, C = \text{concat}(C_1, \dots, C_n) \in \mathbb{R}^{\text{columns: } K_1, \text{ rows: } n \cdot \text{dim}(C_1)}, O = C W^T \in \mathbb{R}^{\text{columns: } K_1, \text{ rows: } n \cdot \text{dim}(C_1)}, X_T = \text{concat}(X_1, \dots, X_n)$$

$$\text{[Satz 2.2]}: \left(f: \gamma \rightarrow \gamma \right) \left(\int \int_{\gamma_1 \times \gamma_2} \left(\int_{\gamma_1} \int_{\gamma_2} f(x,y) dx dy \right) \right), \quad W_1 \in \mathbb{R}^{n \times (n_1 \times n_2)}, \quad W_2 \in \mathbb{R}^{n \times (n_1 \times n_2)}, \quad F = R_W (X_1 V_1^T + b_1) W_2^T + b_2, \quad \text{Aussagen: } L(f, F)$$

PeCodeT

Multi-Head Self Attention

$$Q = X_E W^Q, K = X_E W^K, V = X_E W^V, \in \mathbb{R}^{tokens \times d_{model}}$$

$$\downarrow \text{SFFPE}(h=2)$$

$$\begin{matrix} \text{head 1} & \text{head 2} \\ Q_1, K_1, V_1 & Q_2, K_2, V_2 \\ \in \mathbb{R}^{tokens \times d_k} & \in \mathbb{R}^{tokens \times d_k} \end{matrix}$$

$$S_i = \frac{Q_i K_i^T}{\sqrt{d_k}}, \tilde{S}_i = S_i + M$$

$$M = \begin{bmatrix} 0 & -\infty & -\infty & -\infty & -\infty & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty & -\infty & -\infty & -\infty & -\infty \\ 0 & 0 & 0 & -\infty & -\infty & -\infty & -\infty & -\infty \\ 0 & 0 & 0 & 0 & -\infty & -\infty & -\infty & -\infty \\ 0 & 0 & 0 & 0 & 0 & -\infty & -\infty & -\infty \\ 0 & 0 & 0 & 0 & 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{A}_i = \text{Softmax}(\tilde{S}_i), O_i = \hat{A}_i V_i, \text{Concat}(\text{Concat}(o_1, o_2, \dots)), O_{model} = \text{Concat}(w_{model}^O), Y_1 = \text{LogSoftMax}(X_E + O_{model})$$

Cross Attention

$$Q = Y_1 W^Q, K = X_{model} W^K, V = X_{model} W^V, \in \mathbb{R}^{tokens \times d_{model}}$$

$$\downarrow \text{SFFPE}(h=2)$$

$$\begin{matrix} \text{head 1} & \text{head 2} \\ Q_1^c, K_1^c, V_1^c & Q_2^c, K_2^c, V_2^c \\ \in \mathbb{R}^{tokens \times d_k} & \in \mathbb{R}^{tokens \times d_k} \end{matrix}$$

$$S_i^c = \frac{Q_i^c K_i^{cT}}{\sqrt{d_k}}, \hat{A}_i^c = \text{Softmax}(S_i^c), O_i^c = \hat{A}_i^c V_i^c, C_{model} = \text{Concat}(o_1^c, o_2^c, \dots), O_{model} = \text{Concat}(w_{model}^O), Y_2 = \text{LogSoftMax}(Y_1 + O_{model})$$

FFN

$$(f \rightarrow g \rightarrow f) \int_{w_1}^{d_{model}} \int_{w_2}^{d_{model}} \int_{w_3}^{d_{model}}, W_1 \in \mathbb{R}^{tokens \times tokens}, W_2 \in \mathbb{R}^{tokens \times tokens}, F = \text{ReLU}(Y_2 W_1^T + b_1) W_2^T + b_2, H_{model} = \text{LogSoftMax}(Y_2 + F)$$

Output Projection

$$W_{vocab} \in \mathbb{R}^{vocab_size \times d_{model}}$$

$$Z = H_{model} W_{vocab}^T + b \in \mathbb{R}^{tokens \times vocab_size}$$

$$p = \text{Softmax}(Z)$$

Loss

$$\text{Cross Entropy Loss} (\text{Label Smoothing} = 0.1)$$

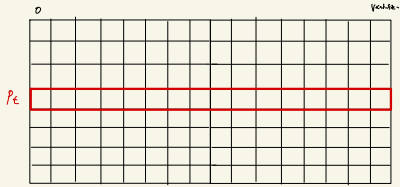
$$Y_{smooth} = (1-f) Y_{model} + \frac{f}{V}$$

$$\text{Output shape} = [B, T, V] \Rightarrow [B, T, V]$$

$$p_i = [\quad], Y_i = [\quad]$$

$$\text{Loss} = - \frac{1}{BT} \sum_{i=0}^{BT-1} \sum_{j=0}^{vocab_size-1} Y_{ij} \log(p_{i,j})$$

Backpropagation



$\in \mathbb{R}^{\text{Embedding} \times \text{Vocab Size}}$

$$Q_t = \text{Embedding} \times \text{Vocab Size}$$

$$P_{t,j} = \frac{e^{z_{t,j}}}{\sum_{k=0}^{\text{vocab}-1} e^{z_{t,k}}}, \quad L_t = - \sum_{j=0}^{\text{vocab}-1} Q_{t,j} \log(P_{t,j})$$

① $\frac{\partial L_t}{\partial z_{t,j}}$

$$P_{t,j} = \frac{e^{z_{t,j}}}{\sum_k e^{z_{t,k}}}, \quad \log P_{t,j} = \log(e^{z_{t,j}}) - \log(\sum_k e^{z_{t,k}}) = z_{t,j} - \log(\sum_k e^{z_{t,k}})$$

$$L = - \sum_j Q_{t,j} \log(P_{t,j}) = - \sum_j Q_{t,j} [z_{t,j} - \log(\sum_k e^{z_{t,k}})] = - \sum_j Q_{t,j} z_{t,j} + \sum_j Q_{t,j} \log(\sum_k e^{z_{t,k}}) = - \sum_j Q_{t,j} z_{t,j} + (\sum_j Q_{t,j}) (\log(\sum_k e^{z_{t,k}}))$$

$$= - \sum_j Q_{t,j} z_{t,j} + \log(\sum_k e^{z_{t,k}})$$

$$L = - \sum_j Q_{t,j} z_{t,j} + \log(\sum_k e^{z_{t,k}})$$

$$\frac{\partial L}{\partial z_{t,j}} = \frac{\partial}{\partial z_{t,j}} [- \sum_j Q_{t,j} z_{t,j} + \log(\sum_k e^{z_{t,k}})]$$

$$= - Q_{t,j} + \frac{\partial}{\partial z_{t,j}} [\log(\sum_k e^{z_{t,k}})]$$

$$\frac{\partial}{\partial z_{t,j}} \log(\sum_k e^{z_{t,k}}) = \frac{1}{\sum_k e^{z_{t,k}}} e^{z_{t,j}} = P_{t,j}$$

$$\frac{\partial L}{\partial z_{t,j}} = P_{t,j} - Q_{t,j}, \quad \frac{\partial L_t}{\partial z_{t,j}} = P_{t,j} - Q_{t,j}, \quad L = \frac{1}{8} \sum L_t, \quad \frac{\partial L}{\partial z_{t,j}} = \frac{1}{8} \frac{\partial L_t}{\partial z_{t,j}} = \frac{1}{8} (P_{t,j} - Q_{t,j})$$

• $\frac{\partial L}{\partial z_{t,j}} = \frac{P_{t,j} - Q_{t,j}}{8} = \frac{P_{t,j} - Q_{t,j}}{\text{Embedding Size}} = G_{t,j}, \quad G = \frac{P - Q}{\text{Embedding Size}} \in \mathbb{R}^{\text{Embedding} \times V}$

② for $W_{vocab} \in \mathbb{R}^{V \times \text{Embedding}}$, $\frac{\partial L}{\partial W_{j,k}} = \frac{1}{8} \sum_{t=0}^7 \frac{\partial L_t}{\partial W_{j,k}} = \frac{1}{8} \sum_{t=0}^7 \frac{\partial L_t}{\partial z_{t,j}} \cdot \frac{\partial z_{t,j}}{\partial W_{j,k}} = \sum_{t=0}^7 \frac{\partial L}{\partial z_{t,j}} \cdot \frac{\partial z_{t,j}}{\partial W_{j,k}}$

$$z = H_{\text{vocab}} W_{\text{vocab}}^T + b, \quad z = \begin{bmatrix} z_{0,j} \\ \vdots \\ z_{7,j} \end{bmatrix} \in \mathbb{R}^{\text{Embedding} \times V}, \quad z_{t,j} = H_{t,k} W_{j,k} + b_j = \sum_{k \in \text{vocab}} H_{t,k} W_{j,k} + b_j$$

$$\frac{\partial L}{\partial W_{j,k}} = \sum_{t=0}^7 \frac{\partial L}{\partial z_{t,j}} \cdot \frac{\partial z_{t,j}}{\partial W_{j,k}} = \sum_{t=0}^7 G_{t,j} H_{t,k}, \quad \begin{bmatrix} & j \\ & \vdots \\ & \end{bmatrix} \quad \begin{bmatrix} & k \\ & \vdots \\ & \end{bmatrix}$$

• $\frac{\partial L}{\partial W_{vocab}} = G^T H$

$$\frac{\partial L}{\partial b_j} = \sum_{t=0}^7 G_{t,j}, \quad \frac{\partial L}{\partial b} = \sum_{t=0}^7 G_t \in \mathbb{R}^{1 \times V}$$

t: target 위치, 0-7
j: vocabulary token, 0-15
k: hidden dimension, 0-3

$$\frac{\partial z_{t,j}}{\partial W_{j,k}} = H_{t,k}$$